

Software Test Plan (STP) - Task Manager Application

Project: Task Manager Application

Version: 1.0

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Date: 17-09-2026

Status: Draft

1. Introduction

Purpose: This document defines the test plan for the Task Manager Application v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers user registration and authentication, task creation and lifecycle management, categorization and search, task assignment and collaboration, notifications and reminders, and reporting/dashboard features. Third-party calendar/email provider integration and payment/billing functionality are excluded, as they are out of scope for the current version.

References: Task Manager Application SRS v1.0 (dated 02-09-2026, Team 6), Design Specifications (referenced per module in the RTM), WCAG 2.1 AA Guidelines.

Definitions: UI (User Interface), API (Application Programming Interface), CRUD (Create, Read, Update, Delete), JWT (JSON Web Token), RBAC (Role-Based Access Control), TLS (Transport Layer Security), NFR (Non-Functional Requirement), RTM (Requirements Traceability Matrix), WCAG (Web Content Accessibility Guidelines), SRS (Software Requirements Specification).

2. Test Items

- Authentication & Account Management module
- Task Creation & Lifecycle Management module
- Collaboration & Sharing module
- Notifications & Reminders module
- Reporting & Dashboard module
- Administration & Maintenance module

3. Features to be Tested

Features mapped to SRS requirement IDs:

4.1 Authentication & Account Management

- TM-F-001: User registration with email and password
- TM-F-002: User authentication (login) via email and password
- TM-F-003: Account lockout after 5 consecutive failed login attempts
- TM-F-004: Forgotten-password reset via time-limited email link

4.2 Task Creation & Lifecycle Management

- TM-F-010: Create a task with title, description, due date, priority
- TM-F-011: Edit title/description/due date/priority/status of an owned task
- TM-F-012: Delete an owned task with confirmation prompt
- TM-F-013: Mark a task complete / reopen a completed task
- TM-F-014: Organize tasks using categories or custom tags
- TM-F-015: Search and filter tasks by title, status, due date, priority, or tag
- TM-F-016: Sort task list by due date, priority, or creation date

4.3 Collaboration & Sharing

- TM-F-020: Assign a task to another registered user (teammate)
- TM-F-021: Add comments to a shared or assigned task
- TM-F-022: Share a task list with view or edit permission

4.4 Notifications & Reminders

- TM-F-030: Reminder notification before due date
- TM-F-031: Notification to teammate on task assignment

4.5 Reporting & Dashboard

- TM-F-040: Dashboard summarizing task counts by status
- TM-F-041: Export task list as CSV or PDF report

4.6 Administration & Maintenance

- TM-F-050: Admin interface to view, disable, or delete user accounts
- TM-F-051: Audit logging of all administrative actions

5. Non-Functional Requirements

- TM-NF-001: Response time $\leq 2s$ (95th percentile, up to 500 concurrent users)
- TM-NF-002: 99.5% monthly uptime
- TM-NF-003: Scalability to 10,000 users / 1,000,000 tasks without degradation
- TM-NF-004: WCAG 2.1 AA accessibility compliance
- TM-NF-005: 30-day recoverable retention of deleted tasks
- TM-NF-006: Responsive usability on desktop and mobile ($\geq 360px$ width)

5.1.2 Security Requirements

- TM-SR-001: TLS 1.2+ enforced on all client-server communications
- TM-SR-002: Salted password hashing (bcrypt/Argon2); no plaintext storage
- TM-SR-003: Role-based access control (RBAC) on task access/edit/delete
- TM-SR-004: JWT session token issuance, expiry, and invalidation on logout
- TM-SR-005: Logging and alerting on repeated failed authentication attempts

4. Features Not to be Tested

- Third-party calendar/email provider integrations
- Payment/billing functionality (out of scope for current version)
- External Notification/Email Service internal implementation (assumed tested by vendor)

5. Test Approach / Strategy

- **Levels:** Unit tests (module-level: AuthModule, TaskModule, CollabModule, NotifModule, ReportModule, AdminModule)

- Integration tests (Client UI ↔ Backend REST API ↔ Database, and ↔ External Notification/Email Service)
- System tests (end-to-end Task Manager Application functionality)
- Acceptance tests (UAT)
- **Types:** Functional testing (core features per module)
- Regression testing
- Performance testing (response time, load, scalability)
- Usability/Accessibility testing (UI clarity, WCAG 2.1 AA compliance)

Entry Criteria: Stable build delivered, test data available (dummy user accounts and tasks), test environment ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all acceptance criteria satisfied.

5.1 Security Validation

- Validate password hashing (bcrypt/Argon2) and confirm no plaintext password storage
- TLS 1.2+ verification for all client-server and server-to-service communication
- Role-based access control (RBAC) checks – verify 403/Forbidden on unauthorized task access
- JWT session token expiry and invalidation-on-logout verification
- Brute-force detection: logout after 5 failed attempts within 10 minutes and audit log/alert generation
- Fuzzing for input fields (registration, task fields, login)
- Penetration testing of authentication and authorization flows

6. Test Environment

Hardware: Standard client devices – desktop, laptop, tablet, smartphone – with internet connectivity; no specialized hardware required.

Software: Task Manager Application v1.0 (web/mobile client), backend REST API, relational/NoSQL database, external Notification/Email Service (sandbox/mock).

Tools: Selenium (UI automation), Postman (API testing), JMeter (performance/load testing), Jira (defect tracking), axe/WAVE (accessibility auditing).

Test Data: Dummy user accounts, sample tasks/categories/tags, mock teammate accounts for collaboration testing.

7. Test Schedule

- **Milestones:** Test case design: 22-Sep-2026
- Environment setup: 24-Sep-2026
- Test execution start: 25-Sep-2026
- Test execution end: 08-Oct-2026
- UAT: 10-Oct-2026 to 13-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports

- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Arushi Punhani	Prepare plan, coordinate execution
Test Engineer	Arnav Sharma	Design & execute test cases, log defects
Developer / Team Lead	Aryan Burman	Support defect fixes and triage, coordinate development
Product Owner	Arya M G	Approve test results, sign-off readiness
Course Coordinator	Asha Kumari	Review and approve overall project deliverables

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from dev team
Test environment downtime	Maintain backup environment on cloud VM
External Notification/Email Service unavailability	Use mock/sandbox notification service for test execution
Data at scale (10,000 users / 1,000,000 tasks) not reproducible in test env	Use load-generation scripts and scaled-down proportional datasets validated against NFR thresholds

11. Assumptions & Dependencies

- Backend REST API and database will be stable and available for testing
- Test data (user accounts, tasks) will be provided/seeded before execution
- External Notification/Email Service sandbox will be available for notification testing

12. Suspension & Resumption Criteria

- **Suspend testing if:** Environment unavailable for >4 hours
- Build is too unstable (blocks >30% of test cases)
- **Resume testing if:** Blocking defects are resolved
- Environment stabilized

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases. The full Requirements Traceability Matrix, covering every functional, non-functional, and security requirement in the SRS, is provided below.

Req ID	Requirement (short)	Section Ref / Design Spec	Module	Test Case(s)	Status
TM-F-001	User registration	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N
TM-F-002	User login	4.1 / DS-Auth-02	AuthModule	TC-Auth-02	N
TM-F-003	Account lockout on failed logins	4.1 / DS-Auth-03	AuthModule	TC-Auth-03	N
TM-F-004	Password reset via email link	4.1 / DS-Auth-04	AuthModule	TC-Auth-04	N
TM-F-010	Create task	4.2 / DS-Task-01	TaskModule	TC-Task-01	N

Req ID	Requirement (short)	Section Ref / Design Spec	Module	Test Case(s)	Status
TM-F-011	Edit task	4.2 / DS-Task-02	TaskModule	TC-Task-02	N
TM-F-012	Delete task with confirmation	4.2 / DS-Task-03	TaskModule	TC-Task-03	N
TM-F-013	Mark task complete / reopen	4.2 / DS-Task-04	TaskModule	TC-Task-04	N
TM-F-014	Categorize / tag tasks	4.2 / DS-Task-05	TaskModule	TC-Task-05	N
TM-F-015	Search & filter tasks	4.2 / DS-Task-06	TaskModule	TC-Task-06	N
TM-F-016	Sort task list	4.2 / DS-Task-07	TaskModule	TC-Task-07	N
TM-F-020	Assign task	4.3 / DS-Collab-01	CollabModule	TC-Collab-01	N
TM-F-021	Add comments to shared task	4.3 / DS-Collab-02	CollabModule	TC-Collab-02	N
TM-F-022	Share task list with permissions	4.3 / DS-Collab-03	CollabModule	TC-Collab-03	N
TM-F-030	Due-date reminder	4.4 / DS-Notif-01	NotifModule	TC-Notif-01	N
TM-F-031	Assignment notification	4.4 / DS-Notif-02	NotifModule	TC-Notif-02	N
TM-F-040	Dashboard summary	4.5 / DS-Report-01	ReportModule	TC-Report-01	N
TM-F-041	Export task list (CSV/PDF)	4.5 / DS-Report-02	ReportModule	TC-Report-02	N
TM-F-050	Admin user management	4.6 / DS-Admin-01	AdminModule	TC-Admin-01	N
TM-F-051	Admin action audit logging	4.6 / DS-Admin-02	AdminModule	TC-Admin-02	N
TM-NF-001	Response time target	5 / DS-Perf-01	WebUI / API	TC-Perf-01	N
TM-NF-002	Uptime target	5 / DS-Ops-01	Infra	Ops reports	N
TM-NF-003	Scalability target	5 / DS-Scale-01	WebUI / API / DB	TC-Scale-01	N
TM-NF-004	WCAG 2.1 AA accessibility	5 / DS-UX-01	WebUI	TC-UX-01	N
TM-NF-005	Deleted-task recoverability (30 days)	5 / DS-Data-01	TaskModule / DB	TC-Data-01	N
TM-NF-006	Responsive UI (desktop & mobile)	5 / DS-UX-02	WebUI	TC-UX-02	N
TM-SR-001	TLS 1.2+ enforcement	5.1.2 / DS-Sec-01	Infra / API	TC-Sec-01	N
TM-SR-002	Password hashing	5.1.2 / DS-Sec-02	AuthModule	TC-Sec-02	N
TM-SR-003	Role-based access control	5.1.2 / DS-Sec-03	AuthModule / TaskModule	TC-Sec-03	N
TM-SR-004	JWT session token expiry/invalidation	5.1.2 / DS-Sec-04	AuthModule	TC-Sec-04	N
TM-SR-005	Brute-force detection & alerting	5.1.2 / DS-Sec-05	AuthModule	TC-Sec-05	N

14. Test Metrics & Reporting

- **Metrics collected:** % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage
- **Reports:** Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Arushi Punhani	Arushi Punhani / 17-09-2026
Dev Lead / Team Lead	Aryan Burman	Aryan Burman / 17-09-2026
Product Owner	Arya M G	Arya M G / 17-09-2026
Course Coordinator	Asha Kumari	Asha Kumari / 17-09-2026